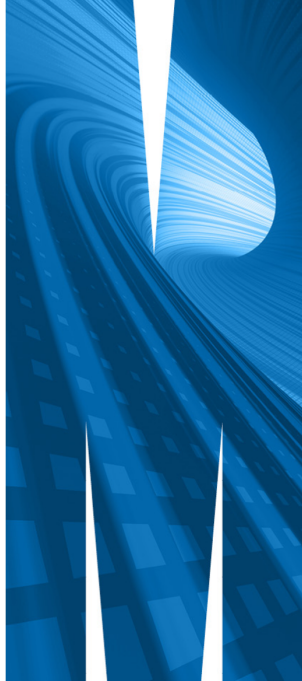


ETC4500/ETC5450

Advanced R programming

Week 4: R programming with LLMs



Outline

- 1 Terminology
- 2 Positron Assistant
- 3 Other tools

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Terminology

- **Provider:** Service that hosts LLM and provides API for interacting with it (e.g., Anthropic)
- **Model:** Specific version of LLM (e.g., Claude-Sonnet-4.6)
- **Prompt:** Input you give LLM to generate a response.
- **System prompt:** Special type of prompt that sets behaviour of LLM.
- **Tokens:** Units of text that the LLM processes (words, parts of words, punctuation, etc.).
- **Context:** Information available to LLM when generating a response
- **Tool:** Function or API that LLM can call to perform specific tasks

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Set up Positron

GitHub Copilot

- Make sure you are logged into your GitHub account.
- GitHub Copilot settings:
<https://github.com/settings/copilot/features>

Anthropic Claude

- Set up Claude as a model provider using an Anthropic API key: <https://platform.claude.com/>
- Ctrl-Shift-P: “Positron Assistant: Configure Language Model Providers”

Code completion

GitHub Copilot provides code completion suggestions as you type.

You can accept suggestions by pressing Tab or Enter

Snooze it by clicking the icon at bottom right of the editor, or by pressing `Ctrl+Shift+P` and selecting “GitHub Copilot: Toggle”

Try typing a function name or a comment describing what you want to do, and see what suggestions Copilot gives you.

Default chat agents

- **Ask:** Provides answers and code suggestions without automatic execution
- **Edit:** Suggests code changes for you to apply to files
- **Agent:** Autonomously executes work including running code and modifying files
- **Plan:** Creates detailed step-by-step plans before execution

You will need a model. Default is GPT-5 mini (slower and less capable than a paid model).

Pick model

- GitHub Copilot: Code completion and chat agents
- Claude: Chat agents with more detailed reasoning and planning capabilities

Chat agent: Ask

- How do I plot a histogram of the `value` variable in my dataset `my_data`?
- Why am I getting a syntax error in my R script?
- Can you explain what this function does?

Attach context

- Default context: active editor file, and active interpreter session.
- Add any other context you think will help.
- Interpreter session context includes code inputs and outputs in that session, in-memory objects in the Variables pane, the active plot in the Plots pane.

Chat agent: Edit

- Refactor this function to be more efficient.
- Add error handling to this code block.
- Update this script to use the latest version of the `dplyr` package.

Chat agent: Agent

- Create a new R script that loads my dataset, performs some basic data cleaning, and generates a summary report.
- Install the required packages and run this analysis script for me.
- Update all my ggplot graphics to use a colour-blind friendly palette.

Chat agent: Plan

- Outline the steps needed to turn this script into an R package.
- Plan how to add tests for this R script
- Create a step-by-step plan to optimise this data processing pipeline.

- You can ask the agent to generate commit messages for you, or even create commits and push them to GitHub.
- You can ask the agent to review your code and suggest improvements, or even automatically apply those improvements.

Slash commands

- `/fix` - Fixes code
- `/explain` - Explains code
- `/exportQuarto` - Generates a Quarto document from the current chat conversation

Prompting tips

- Be specific about what you want the agent to do.
- Provide relevant context to help the agent understand your request.
- Review the agent's suggestions carefully before accepting them, especially if they involve code changes or execution.
- Use the planning agent for complex tasks that require multiple steps or reasoning.

Exercise

Download the `fizzbuzz.R` script from <https://arp.numbat.space/week4/fizzbuzz.R>.

- Ask the Edit agent to update the `fizzbuzz()` function to allow numbers other than 5 and 7 to be used.
- Use the Edit agent to produce roxygen2 documentation for the `fizzbuzz()` function.
- Ask the Plan agent to outline the steps needed to turn the `fizzbuzz()` function into an R package, including adding documentation and tests.

Access Positron Assistant outside panel

- Right-click (or Ctrl-I) in the editor or terminal.
- Highlight code in any editor and click the icon (or Ctrl-.)
- Click “Fix” or “Explain” in the console after an error occurs.

Exercise

Download the `wrangle.R` script from <https://arp.numbat.space/week4/wrangle.R>.

- Select the entire `wrangle()` function and ask: “Explain what this function does, step by step.”
- Select the body of the `wrangle()` function and ask: “What does the `flag` column in `result` represent? What statistical method is being used?”
- Switch to the `Edit` agent and ask: “Add a `roxygen2` documentation block to this function”
- Switch to the `Agent` agent and ask “Write a test that tests that `wrangle()` correctly flags outliers in a known dataset.”

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Other tools

- **Databot:** a Positron extension that provides data analysis capabilities using LLMs, based on the same set up as Positron Assistant.
- **ellmer:** An R package to interact with LLMs from R, including prompt templates and agent frameworks.
- **shinychat** : Build your own custom chat interface using Shiny, and connect it to an LLM using the ellmer package.
- **Claude code:** Terminal-based interface for interacting with Claude, including tools and agent capabilities.

- Interactive data analysis tool that allows you to ask questions about your data and get answers in the form of code, visualizations, and explanations.
- Uses LLMs to generate code and explanations based on your questions.
- Can write Quarto reports based on your data and analysis.

Exercise

- 1 Install the Databot extension from the Positron Assistant marketplace.
- 2 Use the following code to load the gapminder dataset into your workspace.

```
library(gapminder)  
data(gapminder)
```

- 3 Use Databot to look at the relationship between life expectancy and GDP per capita over time. Which countries are outliers?
- 4 Turn the analysis into a Quarto report using /report.

ellmer package

- Allows you to interact with LLMs programmatically from R.
- Beware of long conversations. The entire conversation history is sent with each request.
- Allows a system prompt to set the behaviour of the agent.
- Allows tools to be registered.
- Provides a framework for building custom agents